

From Risk to Wealth: Entrepreneurial Portfolios and Their Economic Consequences*

Sebastian Dyrda

University of Toronto

Baxter Robinson

University of Western Ontario

Renato Zimmermann

University of Toronto

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Abstract

This paper studies how entrepreneur's income risk affects their wealth accumulation and portfolio allocation. We document novel facts about the portfolio allocation of entrepreneurs, showing that entrepreneurs hold substantial financial assets outside their business. We then build a life-cycle model of entrepreneurship that features risky intangible capital. The risky nature of intangible capital helps our model match the portfolio allocation of entrepreneurs of assets inside vs. outside their business. We demonstrate the aggregate importance of this mechanism by studying the aggregate cost of financial frictions in a model with and without intangible capital. We also study two policy applications: an expansion of small business loans and a fiscal stimulus transfer showing that the effects of both policies depend critically on the presence or absence of risky intangible capital.

Keywords: Entrepreneurship, Wealth Inequality, Portfolio Allocation, Intangible Capital, Risk, Aggregate Productivity

JEL Codes: E22, D31, G32, L26

*Dyrda: University of Toronto, sebastian.dyrda@utoronto.ca; Robinson: University of Western Ontario, brobin63@uwo.ca; Zimmermann: University of Toronto, renato.zimmermann@mail.utoronto.ca. This research was enabled in part by support provided by [Compute Ontario](#) and the [Digital Research Alliance of Canada](#). This paper draws on research supported by the Social Sciences and Humanities Research Council.

1 Introduction

Entrepreneurs are disproportionately wealthy. The 12% of U.S. households that include at least one entrepreneur own 48% of the nation’s wealth and constitute 63% of the wealthiest 1% of households.¹ Understanding the mechanisms behind entrepreneurial wealth accumulation is essential not only for explaining the concentrated distribution of wealth but also for comprehending broader economic implications of entrepreneurial decisions. Given the critical role entrepreneurs play in hiring workers and investing in capital, these decisions have significant aggregate implications.

In this paper, we investigate how entrepreneurial income risk affects both the savings behavior and portfolio allocation of entrepreneurs. Specifically, we examine the allocation of wealth between entrepreneur’s investments in their own business and their holding of financial assets outside of their business. Using data from the Survey of Consumer Finances, we document that many entrepreneurial households maintain substantial financial assets outside their businesses, a fact that traditional models of entrepreneurship often overlook.

It is well established that entrepreneurs exhibit higher wealth-to-income ratios than non-entrepreneurs (Quadrini, 1999). The conventional explanation attributes this to borrowing constraints, which ostensibly compel entrepreneurs to save more than workers (Cagetti and De Nardi, 2006). However, we demonstrate that borrowing constraints alone cannot fully account for the observed levels of wealth inequality or the distribution of wealth outside of business holdings. Instead, we argue that adding risky intangible capital in a model of entrepreneurship with borrowing constraints can more accurately reproduce these empirical patterns. This addition reflects the reality that entrepreneurs not only invest in physical capital but also in intangible assets, also known as sweat equity, that are inherently risky yet crucial for sustaining their businesses in a competitive environment.

In our model, intangible capital is risky because it is subject to destruction shocks. These shocks, which occur with an exogenous probability, can reduce the value of intangible capital to zero. This mechanism reflects the fact that due to competitive pressures, some business models periodically become obsolete. Building new intangible capital requires both effort and financial investment. The threat of an intangible capital destruction shock incentivizes entrepreneurs to save beyond what is necessary for tangible capital investment. This heightened savings motive helps explain

¹Authors’ calculations from the 2019 Survey of Consumer Finances.

not only the elevated levels of wealth that entrepreneurs accumulate but also the substantial portion of their wealth that is invested outside of their businesses.

In addition to incorporating risky intangible capital, we introduce Epstein–Zin preferences into our framework, a non-standard feature in the entrepreneurial literature. These preferences allow us to clearly distinguish between the effects of intertemporal substitution and risk—a distinction that is fundamental to understanding portfolio allocation decisions. Together, these features create a new and arguably unique channel through which entrepreneurial risk affects portfolio choices, and further the macroeconomic aggregates. This approach provides a deeper understanding of how entrepreneurial behavior, shaped by both risk and time preferences, impacts broader economic outcomes.

We then proceed to a quantitative analysis by calibrating the model to match key empirical observations. Specifically, we discipline the model’s parameters using three sets of statistics. First, we ensure that the model accurately represents basic macroeconomic aggregates. Second, we calibrate the model to capture realistic income risks faced by both workers and entrepreneurs, particularly replicating the fact that entrepreneurs encounter significantly higher income risks. Third, we align the model with the observed relative income and wealth positions of workers and entrepreneurs, as well as the degree of overall wealth inequality. This calibrated model serves as our measurement tool for the quantitative analysis.

We illustrate the aggregate implications of our mechanism by examining the macroeconomic costs of financial frictions in a model with and without risky intangible capital. By relaxing the borrowing constraints that entrepreneurs face, capital is reallocated from wealthy entrepreneurs to those with higher ability. In a model without risky intangible capital, wealth inequality is primarily driven by borrowing constraints, and a substantial number of entrepreneurs are financially constrained. In such an economy, relaxing these constraints leads to significant increases in aggregate output and productivity. However, in our model with risky intangible capital, which better matches the distribution of entrepreneurial portfolios, far fewer entrepreneurs are constrained. Consequently, the aggregate cost of financial frictions is considerably lower in this scenario.

We then illustrate that these quantitative results have important policy implications. We first study the U.S. Small Business Administration loan program. We show how the expansion of government provided loans has large effects in an economy without risky intangible capital, and much more modest effects in an economy with risky intangible capital. We then study a fiscal stimulus transfer, and show that the two models imply very different fiscal multipliers.

2 Related Literature

We contribute to three literatures. Firstly, our paper builds on a literature studying the connection between entrepreneurship and wealth. Since at least [Quadrini \(1999\)](#), it has been well known that entrepreneurs account for a large share of aggregate wealth and are disproportionately found at the top of the wealth distribution.

[Cagetti and De Nardi \(2006\)](#) demonstrated that a model of entrepreneurship featuring borrowing constraints can quantitatively explain the distribution of wealth. We contribute by bringing in information about entrepreneur’s portfolio allocation inside vs. outside their businesses and using that to better discipline the degree to which borrowing constraints can generate strong entrepreneurial savings motives.

We also contribute to a literature studying organizational or intangible capital. Notable work includes [Atkeson and Kehoe \(2005\)](#), [Eisfeldt and Papanikolaou \(2014\)](#), [Eisfeldt and Papanikolaou \(2013\)](#), [McGrattan \(2020\)](#), and [Bhandari and McGrattan \(2021\)](#). Here our contribution is to explore the implication of the risky nature of intangible capital for entrepreneur’s portfolio allocation as well as the aggregate implications of this risky nature.

Finally, we contribute to a literature measuring the macroeconomic implications of financial frictions. Papers such as [Buera \(2009\)](#), [Moll \(2014\)](#), [Midrigan and Xu \(2014\)](#), and [Buera, Kaboski, and Shin \(2011\)](#) all study how borrowing constraints distort entrepreneurial decisions through a variety of mechanisms and how that reduces aggregate output or productivity. Notable contributions in this area also include [Banerjee and Duflo \(2005\)](#) who explore how financial constraints limit entrepreneurial activity in developing countries, and [Hurst and Lusardi \(2004\)](#) who investigate the relationship between wealth, liquidity constraints, and entrepreneurship in the U.S. context. Our work adds to this literature by emphasizing the role of risky intangible capital in shaping the portfolio decisions of entrepreneurs and the broader macroeconomic implications of these decisions.

[Robinson \(2021\)](#) compares how missing markets for risk and borrowing constraints distort entrepreneurial decisions and the aggregate impacts of those distorted decisions in a model with endogenous risk. By contrast, our paper studies how a particular savings mechanism: the need to save in response to risky intangible capital changes the implied aggregate losses from borrowing constraints.

3 Empirical Facts

In this section, we document several facts from the 2019 Survey of Consumer Finances (SCF) about the level of wealth and the portfolio allocation of entrepreneurs.

Table 1: Business Owner Wealth

	% of Sample	Average Wealth	Average Income	Wealth-to-Income Ratio	Non-Business Wealth	Non-Business Wealth-To-Income
Full Sample	100	745	104	7.1	578	5.5
Non-Business Owners	89	404	83	4.9	404	4.9
Active Business Owners	11	3,126	255	12.3	1,667	6.5
Passive Business Owners	2	7,915	555	14.3	4,922	8.9
Incorporated Business Owners	1	5,640	353	16.0	2,630	7.5
Unincorporated Business Owners	10	3,016	251	12.0	1,644	6.6

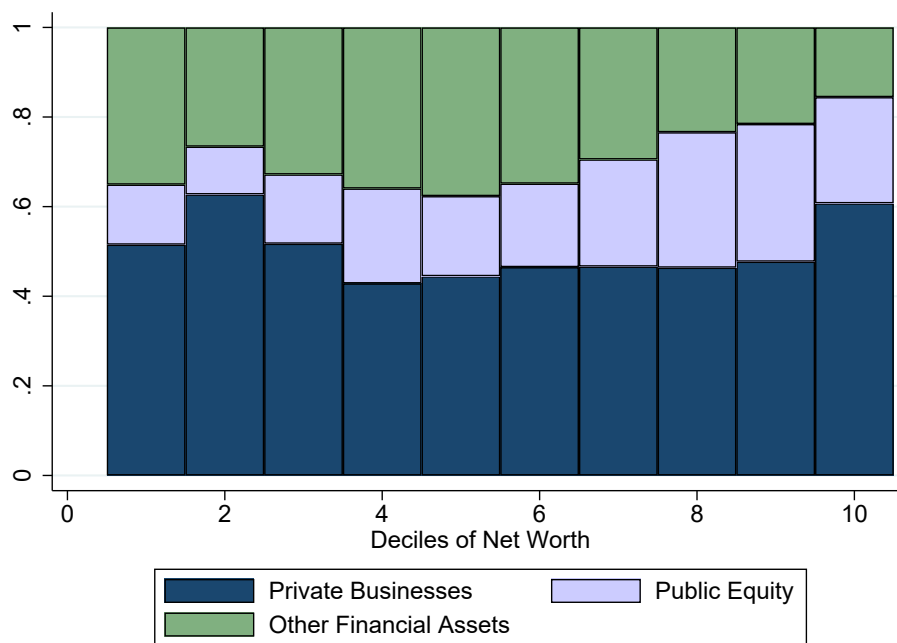
Source: Survey of Consumer Finances (2019). All wealth-to-income ratios are calculated as average group wealth divided by average group income.

Entrepreneurs are much wealthier than non-business owners. We define entrepreneurs as individuals who both own and actively manage a business. Table 1 shows, households with at least entrepreneur have an average wealth of 3.1 million compared to 404 thousand for household that do not own any businesses. Of course, these entrepreneurial households also have much higher income, but their wealth-to-income ratios are also substantially higher than non-business owning households. Entrepreneurial households collectively have a wealth-to-income ratio of 12.3 relative to 4.9 for non-business owning households.

Some of these differences in wealth-to-income ratios are due to the value of the businesses owned by entrepreneurial households. However, even after excluding the value of privately held businesses, these entrepreneurial households own substantially more wealth and have higher wealth-to-income ratios than non-business owning households. Entrepreneurial households have non-business-wealth-to-income ratios of 6.5 relative to non-business owning household's 4.9.

The fact that entrepreneurial households have higher wealth-to-income ratios than non-business owning households, even after excluding business wealth, suggests that entrepreneurs do not invest all of their wealth in their businesses. Figure 1 shows the average portfolio composition of business owning households with different levels of wealth, excluding real estate, vehicles, and other tangible assets. Across the deciles of wealth, the total value of businesses accounts for 40-60% of financial assets.

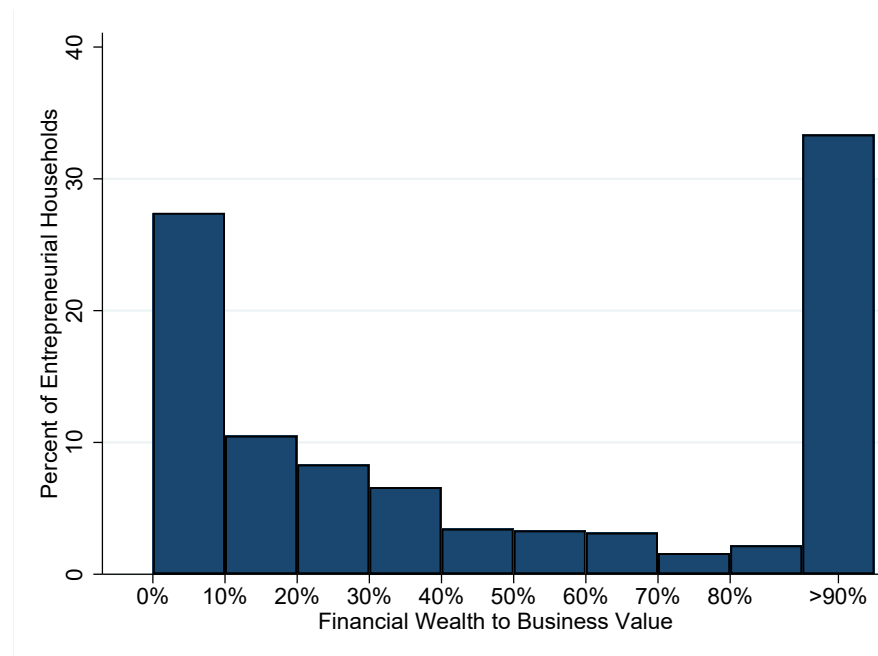
Figure 1: Portfolio Composition of Entrepreneurs



Notes: The portfolio composition of financial assets owned by entrepreneurial households over the distribution of wealth. Deciles of net worth are calculated based on the distribution of wealth for households that own and actively manage at least one business. Real estate, vehicles and other tangible assets are excluded.

This prevalence of wealth invested outside of the business suggests that not all entrepreneurs are borrowing constrained. A entrepreneur who faces a binding borrowing constraint wishes to invest more in their business but is unable to do so because of the borrowing constraint. Presumably they want to invest more because the marginal returns to additional investment is high. As a result, they should want to invest most, if not all, of their available financial assets in their businesses. The prevalence of wealth outside the business is one indication that not all entrepreneurs are borrowing constrained, and we see from figure 1 that there are some entrepreneurs across the distribution of wealth for whom this is true.

Figure 2: Distribution of Financial Wealth Relative to Business Value



Notes: This figure plots the distribution of Entrepreneur's financial wealth held outside the business relative to the value of their business.

A more direct way to assess the proportion of entrepreneurs who might be borrowing constrained is to compare entrepreneur's financial wealth held outside the business to the total business value. Figure 2 shows the distribution of the ratio of financial wealth to business value. 27% of entrepreneurs have less than 10% of the value of their business held in financial assets outside the

business. These entrepreneurs are the most likely to be borrowing constrained, as they have few financial assets outside the business. Another 11% of entrepreneurs have between 10% and 20% of the value of their business held in financial assets outside the business. The remaining 62% of entrepreneurs have more than 20% of the value of their business held as financial assets outside their business. Given this, if these entrepreneurs wanted to expand the scale of their business, they have financial resources they could use.

What do entrepreneurs say are their reasons for savings? Table 2 reports the proportion of entrepreneurial households that mention different savings motives when asked. Only 7% of entrepreneurial households say they are saving in order to invest, again suggesting that the proportion of entrepreneurs who are seeking to save out of their borrowing constraints is limited. By contrast 48% say they are saving for precautionary reasons, and 55% say they are saving for retirement. Note that households are able to give multiple reasons why they are saving and therefore the percentages do not add up to 100%.

Table 2: Self-Reported Savings Motives

	Proportion of Households Saving for:				
	Investment	Precautionary	Retirement	Purchases	Home
Full Sample	3	51	38	22	12
Non-Business Owners	3	51	36	23	13
Active Business Owners	7	48	55	20	5

Notes: This table reports the percentage of households that claim they are saving for five broad reasons. “Investment” includes investing both inside and outside their business. “Precautionary” includes job losses, illness and other emergencies. “Purchases” includes a variety of durable goods, vacations, vehicles, vacation properties, to pay off debt, and gifts. “Home” includes buying a home, improving or repairing a home and moving expenses. Percentages do not add up to 100% as respondents were able to provide multiple answers and not all categories are reported. Source: Survey of Consumer Finances (2019)

4 The Model of Entrepreneurship with Endogenous Portfolios

As in [Lucas \(1978\)](#), firms produce a homogeneous good using a technology characterized by decreasing returns to scale and vary in their productivity. Output is generated using capital and labor sourced from competitive markets, along with intangible capital accumulated by entrepreneurs. We assume that firms face two sources of financial frictions. First, motivated by the evidence in [Moskowitz and Vissing-Jørgensen \(2002\)](#) that entrepreneurial investment is poorly

diversified, we follow [Quadrini \(2000\)](#), [Cagetti and De Nardi \(2006\)](#), and [Buera et al. \(2011\)](#) in assuming that each firm is entirely owned by a single entrepreneur who can only partially insure business income risk by saving in a risk-free asset. Second, we assume a collateral constraint that limits the firm's borrowing capacity. Similar to [Bhandari and McGrattan \(2021\)](#), we assume that entrepreneurs accumulate intangible capital or "sweat equity," which serves as a factor of production.

We extend the macro-entrepreneurship literature by incorporating two additional features into our model. First, we introduce Epstein-Zin preferences, which allows us to clearly separate the impact of the intertemporal substitution channel from the risk channel—a distinction fundamental to portfolio allocation decisions. Second, we introduce "disaster" risk associated with the accumulation of intangible capital. Each period, entrepreneurs face a positive probability of their intangible capital being wiped out, which we interpret as a shorthand for events such as being outcompeted or technology becoming obsolete. The details of the model are presented in the following sections.

4.1 Environment

There are a continuum of households. Households are born at age $j = 1$ and die with certainty at $j = J$. At each age j , an individual has preferences over consumption (c_j) and the disutility of exerting effort (h_j) of:

$$U_t = \left[\left(c_t^\psi (1 - h_t)^{1-\psi} \right)^{\frac{1-\gamma}{\zeta}} + \beta \left(\mathbb{E}_t U_{t+1}^{1-\gamma} \right)^{\frac{1}{\zeta}} \right]^{\frac{\zeta}{1-\gamma}} \quad (1)$$

where $\gamma \geq 0$ is the parameter that controls risk aversion, $\sigma \geq 0$ is the intertemporal elasticity of substitution (IES), and:

$$\zeta = \frac{1 - \gamma}{1 - \frac{1}{\sigma}}.$$

The term $\left(\mathbb{E}_t U_{t+1}^{1-\gamma} \right)^{\frac{1}{1-\gamma}}$ is often called the risk-adjusted expectation operator. When $\gamma = \frac{1}{\sigma}$, $\zeta = 1$, the recursive preferences collapse to the standard case of state-separable, CRRA utility. Epstein-Zin preferences imply that the agents have preferences for early or later resolution of uncertainty. In our notation, if $\gamma > \frac{1}{\sigma}$, they prefer an early resolution of uncertainty, and if $\gamma < \frac{1}{\sigma}$, a later resolution. Finally, the discount factor is β .

A proportion $(1 - \mu)$ of individual are workers and the rest (μ) are entrepreneurs. We describe their decisions in what follows.

4.2 Entrepreneurs

Entrepreneurs is characterized by entrepreneurial ability (z) drawn from the distribution F_z , a level of intangible capital (κ) and assets in terms of the risk-free bond (a). Each period, they choose how much capital to rent (k) and efficiency units of labour to hire (n) and earn profits according to:

$$\pi(z, a, \kappa) = \max_{n, k} z(\kappa^\theta k^\alpha n^{1-\theta-\alpha})^\gamma - r_k k - wn \quad (2)$$

Capital investment k is subject to the following exogenous borrowing constraint:

$$k \leq \lambda(j, z) \times a \quad (3)$$

where $\lambda(j, z) \in [1, \infty)$ is a parameter which determines the tightness of borrowing constraint. We allow it to be potentially dependent on the business's age j and productivity z , but not on intangible capital.²

Entrepreneurs can invest in their level of intangible capital (κ) by exerting effort h and investing financial resources q . At the same time, intangible capital is risky, in that it is subject to occasional destruction shocks. With a probability $(1 - p)$ an entrepreneur's intangible capital is reduced to zero. Intangible capital therefore evolves according to:

$$\kappa' = \begin{cases} (1 - \delta)\kappa + q^\delta h^\epsilon & \text{with probability } p \\ 0 & \text{with probability } 1 - p \end{cases} \quad (4)$$

Each period entrepreneurs choose how much to consume c , how much to save in the bond a' and how much financial resources to invest in growing their stock of intangible capital q . They also choose how much effort to exert in producing new intangible capital h .

²Intangible capital, such as intellectual property, brand equity, and human capital, is generally not collateralizable because it lacks physical form and can be challenging to value and liquidate in the event of default. This non-collateralizability limits its usefulness in securing loans, particularly for small and medium-sized enterprises. See [Eisfeldt and Papanikolaou \(2014\)](#) for a discussion on the valuation and ownership of intangible capital.

Thus, at each age $j < J$, the entrepreneur's value function is given by:

$$V^E(a, \kappa, z, j) = \max_{c_E, h_E, q, a'_E, \kappa'} \left[\left(c_E^\psi (1 - h_E)^{1-\psi} \right)^{\frac{1-\gamma}{\zeta}} + \beta \left(\mathbb{E} V^E(a'_E, \kappa', z', j+1)^{1-\gamma} \right)^{\frac{1}{\zeta}} \right]^{\frac{\zeta}{1-\gamma}} \quad (5)$$

subject to:

$$\begin{aligned} c_E + a' + q &= \pi(z, a, \kappa) + (1 + r^a)a \\ \kappa' &= \begin{cases} (1 - \delta_\kappa)\kappa + q^\vartheta h_E^\epsilon & \text{with probability } p \\ 0 & \text{with probability } 1 - p \end{cases} \\ \pi(z, a, \kappa) &= \max_{n, k} z \kappa^\phi (k^\alpha n^{1-\alpha})^\gamma - r^k k - w n \\ k &\leq \lambda(j, z) a \end{aligned}$$

and where $V^E(a, \kappa, z, J) = 0, \forall z, a, \kappa$.

4.3 Workers

Workers have stochastic labour efficiency (y) and choose how much labour effort to exert (h_W), resulting in labour earnings of $wy h_W$, where w is a wage per efficiency unit of labour. Workers also make a consumption-savings decision each period, saving in a risk-free bond a . At each age j , the worker's value function is given by:

$$V^W(a, y, j) = \max_{c_W, h_W, a', \kappa'} \left[\left(c_W^\psi (1 - h_W)^{1-\psi} \right)^{\frac{1-\gamma}{\zeta}} + \beta \left(\mathbb{E} V^W(a'_E, y', j+1)^{1-\gamma} \right)^{\frac{1}{\zeta}} \right]^{\frac{\zeta}{1-\gamma}} \quad (6)$$

subject to

$$\begin{aligned} c_W + a' &= wy h_W + (1 + r_a)a \\ a' &\geq \underline{a} \end{aligned}$$

and where $V^W(a, y, J) = 0, \forall a, y$.

4.4 Equilibrium

An equilibrium is a set of value functions $\{V^W(a, y, j), V^E(a, \kappa, z, j)\}$ a set of policy functions $\{c_W, h_W, a'_W, c_E, h_E, a'_E, q, \kappa', n, k\}$, a distribution of agents $\{\Gamma^E(j, z, a, \kappa), \Gamma^W(j, y, a)\}$, and a price w such that

1. The policy functions solve the individual's problems given by (5) and (6).

2. Labor market clears:

$$\int y h_W d\Gamma^W(j, y, a) = \int n d\Gamma^E(j, z, a, \kappa)$$

3. Goods market clears:

$$\begin{aligned} \int (c_W + a') d\Gamma^W(j, y, a) + \int (c_E + a' + q) d\Gamma^E(j, z, a, \kappa) = \\ \int \left(z \kappa^\phi (k^\alpha n^{1-\alpha})^\gamma + (1 + r^A) a \right) d\Gamma^E(j, z, a, \kappa) + \int ((1 + r^A) a) d\Gamma^W(j, y, a) \end{aligned}$$

4. The distribution of agents is stationary

$$\Gamma^E(j, z, a, \kappa) = \Gamma^{E'}(j, z, a, \kappa)$$

$$\Gamma^W(j, y, a) = \Gamma^{W'}(j, y, a)$$

Note that at this stage we assume the economy is a small open economy so that the interest rate is set exogenously and the financial asset can be in net positive or negative supply.

5 Taking the Model to the Data

5.1 Calibration

In this section, we describe how we calibrate the model. We split the parameters into two group. For the first group, we use commonly used values from the literature. For the second group, we jointly calibrate their value to target a number of data moments.

Externally Chosen Parameters. A period in the model is a year. We set the capital share α to 0.3 and the interest rate r_a to 4% per year. The cost of capital is the interest rate plus the

depreciation rate $r_k = r_a + \delta_k$. We set the tangible capital depreciation rate δ_k to 6% and the intangible capital depreciation δ_κ to 0%. We set the coefficient of relative risk-aversion to be two.

Table 3: Externally Chosen Parameters

Parameter	Value	Source
β Discount factor	0.95	Standard
σ Coefficient of relative risk-aversion	2.0	Standard
α Capital share	0.25	Standard
δ_k Depreciation of tangible capital	0.06	Standard
μ Proportion of entrepreneurs	0.11	SCF (2019)
ϑ Monetary investment to intangible capital share	0.4	Bhandari and McGrattan (2021)
ϵ Time investment to intangible capital share	0.4	Bhandari and McGrattan (2021)
θ Intangible capital to production share	0.25	Bhandari and McGrattan (2021)
δ_κ Depreciation of intangible capital	0.0	Bhandari and McGrattan (2021)

We calculate the proportion of households with at least one active business owner in the 2019 Survey of Consumer Finances as 12% and use this to discipline μ , the proportion of entrepreneurs in our model. Lacking good data on the value of intangible capital we also set the four parameters that govern intangible capital $(\vartheta, \epsilon, \theta, \delta_\kappa)$ equal to the values used in [Bhandari and McGrattan \(2021\)](#). The externally chosen parameters are reported in Table 3.

Internally Chosen Parameters. We then endogenously calibrate the remaining parameters to match the several data moments.

We use data moments computed by [DeBacker, Panousi, and Ramnath \(2023\)](#) from the American Internal Revenue Service (IRS) to discipline both the workers and entrepreneur’s income processes. Given that our main mechanism has to do with the risky nature of intangible capital, we want the income changes in our model to be consistent with the data. To that end, we target the proportion of entrepreneurs that experience a one-year decline in income of more than 50%, the proportion that experience a one-year change of between -10% to 10% and the proportion that experience a one-year increase of more than 50%. We also target the same three statistics for worker’s one-year income changes. The statistics provided by the [DeBacker et al. \(2023\)](#) rely on pass-through businesses only, potentially raising the issue of how representative this group is for the entire population of businesses. Though, as shown in [Dyrda and Pugsley \(2019\)](#) and [Dyrda and Pugsley \(2024\)](#) the pass-throughs became a dominant form of business organization in the United States over the last four decades.

In order to ensure our model has a realistic degree of wealth inequality we target several statistics from the 2019 version of the Survey of Consumer Finances. Specifically, we target the wealth-to-income ratio of entrepreneurs, the wealth-to-income ratio of workers, the top 1% wealth share, the top 10% wealth share, and the ratio of average entrepreneurial income to average worker income.

In order to discipline the strength of the borrowing constraints, we target the average leverage from [Dinlersoz, Kalemli-Ozcan, Hyatt, and Penciakova \(2019\)](#), who build a matched dataset using the Longitudinal Business Database from the U.S. Census Bureau and both Orbis and Compustat. Finally, we target the capital-output ratio of the economy from Federal Reserve Economic Data (FRED). We report the targeted moments and the model’s performance relative to them in Table 4, and the calibrated parameters in Table 5.

Table 4: Moments: Data vs. Model

Moments	Data	Model	Source
Worker Income Changes < 50%	0.04	0.03	DeBacker, Panousi, Ramnath (2023)
Worker Income Changes $\in [-10, 10]\%$	0.52	0.27	DeBacker, Panousi, Ramnath (2023)
Worker Income Changes > 50%	0.08	0.08	DeBacker, Panousi, Ramnath (2023)
Entrepreneur Income Changes < 50%	0.27	0.19	DeBacker, Panousi, Ramnath (2023)
Entrepreneur Income Changes $\in [-10, 10]\%$	0.14	0.12	DeBacker, Panousi, Ramnath (2023)
Entrepreneur Income Changes > 50%	0.3	0.29	DeBacker, Panousi, Ramnath (2023)
Ratio of Entrepreneur to Worker Income	3.0	5.09	SCF (2019)
Wealth to Income of Entrepreneurs	12.1	7.13	SCF (2019)
Wealth to Income of Workers	4.8	3.03	SCF (2019)
Top 1% Wealth Share	0.37	0.1	SCF (2019)
Top 10% Wealth Share	0.76	0.58	SCF (2019)
Leverage	0.46	0.47	Dinlersoz, Kalemli-Ozcan, et al (2019)
Capital-Output Ratio	2.57	2.49	FRED (2009-2019)

5.2 Validation

We validate our model by comparing the portfolio allocation of entrepreneurs in our calibrated model to the data and to an alternative economy without intangible capital. The results are presented in Figure 3. The left panel shows the ratio of average equity to average wealth among entrepreneurs across the wealth distribution. In the data, this ratio oscillates around 0.5 across the deciles of net worth. While our model does not perfectly replicate this pattern, it does produce an average allocation that is consistent with the data. Importantly, these allocations are not directly targeted in our calibration procedure, which means they serve as an external validation

Table 5: Internally Calibrated Parameters

Parameter		Value
γ	Returns to scale	1.0
μ_z	Average entrepreneurial productivity	0.38
ψ	Weight on effort disutility	1.3
ν	Curvature on effort disutility	0.9
p	Probability of no capital destruction shock	0.72
λ	Tightness of collateral constraint	18.66
σ_w	Standard deviation worker productivity shock	0.21
ρ_w	Autocorrelation of worker productivity shock	1.0
σ_z	Standard deviation entrepreneurial productivity shock	1.48
ρ_z	Autocorrelation of entrepreneurial productivity shock	0.66
w_0^E	Initial Entrepreneurial Wealth	33.96
w_0^W	Initial Worker Wealth	86.89

of our model. It is instructive to compare our model's performance to that of an economy without intangible capital (labeled "No κ " in the figure). We use the same parameter values in this counterfactual economy as in our baseline and examine how well this model accounts for portfolio allocations. As illustrated by the green line in the left panel of Figure 3, this exercise yields counterfactual portfolio allocations in which entrepreneurs would invest all of their wealth in equity, regardless of their decile of net worth. Thus, we infer that risky intangible capital, as modeled, is crucial for accurately accounting for equity allocations.

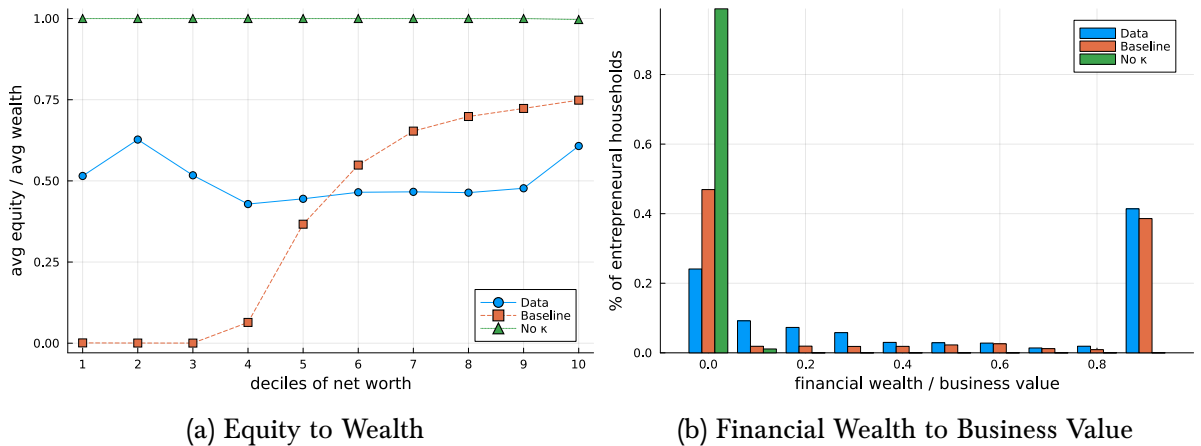


Figure 3: Portfolios of entrepreneurs: Baseline vs. No-intangible capital model

The right panel of Figure 3 offers an alternative illustration of the portfolio allocation of entrepreneurs. It shows the fractions of entrepreneurs with specific ratios of financial wealth to business values. In the data, 20 percent of entrepreneurs hold financial assets worth less than 10% of their businesses value, while around 40 percent have more than 100 percent of their business values invested in financial wealth. Our model successfully replicates this bimodal distribution of asset allocation, despite the fact that this is not explicitly targeted. By contrast, the economy without intangible capital places all the mass of entrepreneurs at zero financial assets. These findings confirm the crucial role that risky intangible capital plays in portfolio allocations.

6 Inspecting the Mechanism

In this section we dig deeper into the key elements determining the economic properties of the model. We begin by inspecting the life-cycle patterns of decisions for entrepreneurs. We then move to illustrate the allocative role of intangible capital focusing on its macroeconomic implications, which allows us to link our findings to those from [Bhandari and McGrattan \(2021\)](#).

6.1 Life-cycle patterns

Figure 4 presents life-cycle profiles of entrepreneurs' average consumption, risk-free assets, effort, and intangible capital across different productivity levels. On average, consumption increases with age, and consistent with empirical evidence, the cross-sectional dispersion of consumption also rises with age.³ In contrast, the behavior of risk-free assets shows the opposite pattern: holdings decline sharply with age, while the dispersion remains relatively stable. Initially, entrepreneurs invest heavily in risk-free assets as this helps to relax their borrowing constraints for tangible capital, allowing them to grow their businesses at a faster rate and capture the high marginal returns on investment through business income. However, as the business matures and the entrepreneur's planning horizon shortens, this incentive diminishes. Absent bequest motives, entrepreneurs end their lives with no assets.

Turning to the life-cycle pattern of effort, entrepreneurs exert high effort early on as they strive to invest in greater intangible capital and extract high marginal returns. As returns diminish with

³The observation that cross-sectional dispersion of consumption increases with age is well-documented in the literature. As individuals age, differences in lifetime income, savings behavior, and health expenses contribute to rising consumption inequality. [Attanasio and Weber \(2010\)](#) provide a comprehensive review of models explaining this phenomenon, while [Aguiar and Hurst \(2013\)](#) empirically show how life-cycle expenditure patterns, including consumption, exhibit increasing dispersion with age in the U.S.

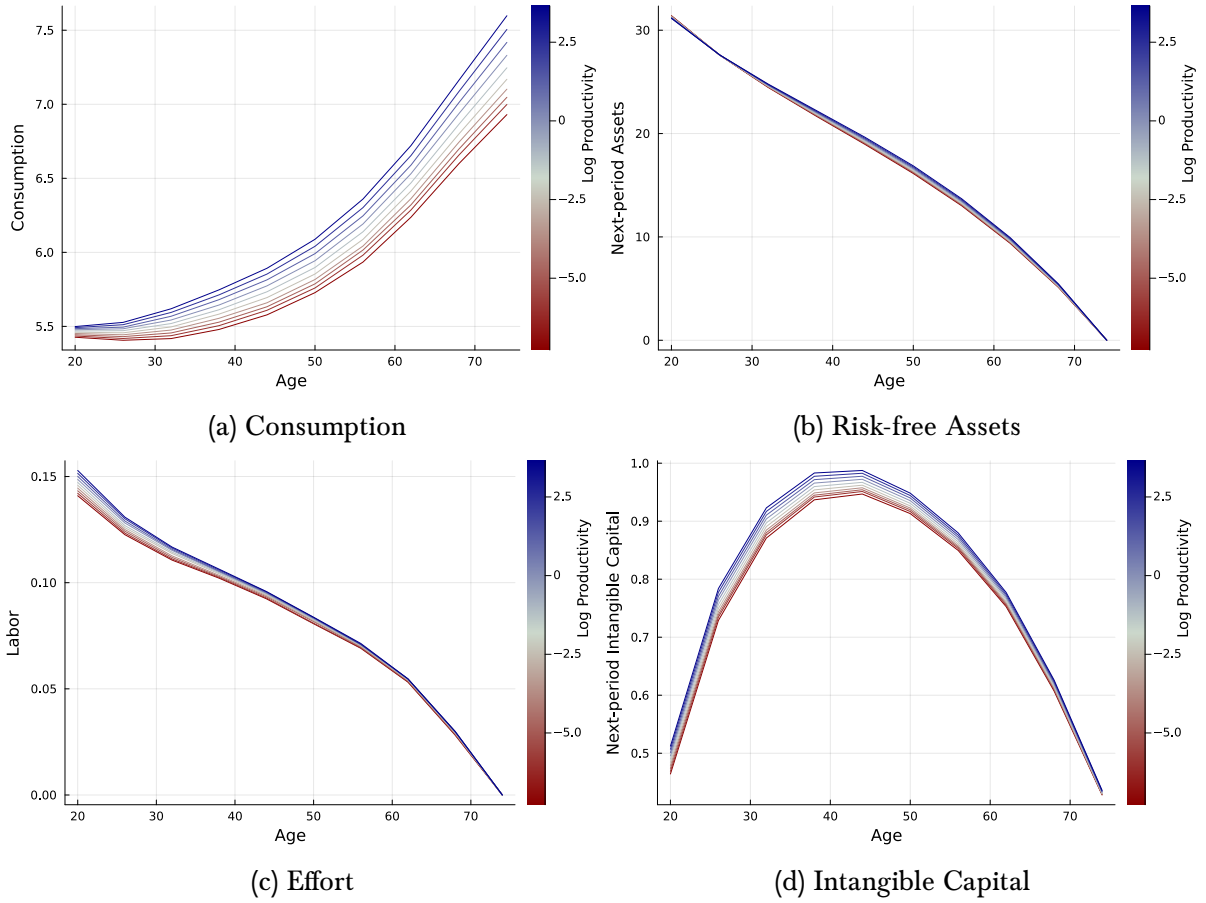


Figure 4: Life-cycle profiles for entrepreneurs

increasing business scale and the number of remaining years declines, effort decreases rapidly, reaching zero in the final period. Unlike consumption, the cross-sectional variation in effort is highest at the beginning of an entrepreneur's lifetime. The pattern of intangible capital over the life-cycle exhibits an inverted U-shape. It takes time for entrepreneurs to accumulate intangible capital, which peaks in mid-life. Over time, as businesses reach maturity and the remaining years decrease, intangible capital declines, though it never falls to zero. This reflects the necessity of supporting a growing consumption pattern over the life-cycle. In contrast to both consumption and effort profiles, the cross-sectional variance in intangible capital is maximized around mid-life for entrepreneurs.

Table 6: Allocative Role of Intangible Capital and Effort

	Baseline	Low Intang. Share ($\theta = 0.125$)	High Intang. Share ($\theta = 0.50$)	Low Effort Share ($\varepsilon = 0.3$)	High Effort Share ($\varepsilon = 0.5$)
GDP	3.158	135.308	0.028	3.951	2.496
Labor	15.7	840.837	0.07	19.641	12.407
Tangible Capital	7.695	82.017	0.07	9.479	6.137
Intangible Capital	0.521	0.545	0.102	0.668	0.405
TFP	0.623	0.728	0.242	0.66	0.587
Wage Bill/GDP	0.497	0.621	0.249	0.497	0.497
Profits/GDP	0.259	0.318	0.503	0.263	0.257
Interest Income/GDP	0.261	0.025	30.34	0.209	0.33

6.2 Allocative role of intangible capital

Table 6 presents a detailed examination of how varying the share of intangible capital (θ) and the share of effort (ε) in the production function impacts key macroeconomic variables in partial equilibrium. The second and third columns of the table explore the effects of altering the share of intangible capital in the production function, while the fourth and fifth columns examine changes in the share of effort. Compared to the baseline scenario (first column), increasing the share of intangible capital (θ) significantly amplifies GDP, labor, and tangible capital. Specifically, when the intangible capital share is increased from 0.25 (baseline) to 0.50, GDP rises markedly, reflecting the enhanced role of intangible assets in driving production. Total factor productivity (TFP) also increases with a higher share of intangible capital, underscoring its critical role in boosting efficiency and output. Conversely, reducing the share of intangible capital to 0.125 diminishes GDP and TFP substantially, highlighting the importance of intangible capital in sustaining economic growth.

Similarly, the variation in the share of effort (ε) in the production of intangible capital has profound macroeconomic implications. When the effort share is increased from 0.4 (baseline) to 0.5, GDP and labor input both increase, indicating that higher entrepreneurial effort in accumulating intangible capital translates into greater overall economic output. This is accompanied by a rise in intangible capital and TFP, suggesting that effort is a crucial driver of productivity in the economy. On the other hand, reducing the effort share to 0.3 results in a significant decline in GDP,

labor, and TFP, signaling that lower effort in intangible capital accumulation constrains economic performance. Across these scenarios, the wage bill to GDP ratio remains relatively stable, while profits as a share of GDP slightly adjust in response to the changes in intangible capital and effort shares, indicating a nuanced redistribution of income between labor and capital based on these parameters.

7 Quantitative Experiment

Our main quantitative experiment is to measure the aggregate losses due to borrowing constraints and how these losses change with the presence or absence of risky intangible capital.

We start with our benchmark model with risky intangible capital. We relax the borrowing constraints by setting $\lambda = \infty$ so that entrepreneurs can invest in any amount of capital (k) regardless of their available financial assets (a). We then compare aggregate output and productivity in the economy with our calibrated borrowing constraints $\lambda < \infty$ to a relaxed borrowing constrained economy $\lambda = \infty$. The results of this experiments are presented in tables 7 and 8.

Table 7 presents the results of a quantitative experiment that compares the aggregate allocations in two different economic scenarios: the baseline economy with risky intangible capital and a counterfactual economy without risky intangible capital. We call the counterfactual economy without risky intangible capital the Cagetti De-Nardi economy (or C-DN Economy for short), as it is similar to the model in the seminal [Cagetti and De Nardi \(2006\)](#)⁴. The table provides insights into key macroeconomic variables, including GDP, labor, tangible capital, intangible capital, total factor productivity (TFP), the wage bill as a share of GDP, profits as a share of GDP, and interest income as a share of GDP.

In the baseline economy, the presence of risky intangible capital results in slightly lower GDP and TFP compared to the C-DN economy. However, the baseline economy shows higher levels of labor, tangible capital, and intangible capital, reflecting the role of intangible capital in supporting higher capital and labor inputs. The wage bill to GDP ratio is marginally higher in the baseline economy, while the profits to GDP ratio is slightly lower, suggesting a redistribution of income between labor and capital. Interest income as a share of GDP remains similar across both scenarios, indicating that the financial sector's role is not significantly altered by the presence of risky intangible capital.

⁴[Cagetti and De Nardi \(2006\)](#) also include a luxury bequest motive, have two-ages, young and old, which individuals probabilistically age through, and individuals make occupational choices in that model.

Table 7: The Role of the Collateral Constraint: Allocations

	Baseline	C-DN Economy	Baseline Unconstrained	C-DN Economy Unconstrained
GDP	0.92	0.71	0.97	0.79
Labor	2.28	1.7	2.59	1.79
Tangible Capital	2.29	1.55	2.41	1.88
Intangible Capital	0.57	N/A	0.56	N/A
TFP	0.65	0.9	0.64	0.93
Wage Bill/GDP	0.5	0.47	0.49	0.47
Profits/GDP	0.25	0.31	0.26	0.29
Interest Income/GDP	0.83	0.83	0.84	0.64

Notes: C-DN Economy denotes Cagetti, De-Nardi economy i.e. economy without risky intangible capital following [Cagetti and De Nardi \(2006\)](#) setup. Unconstrained economies are counterfactuals with no collateral constraints for the entrepreneurs, i.e. $\lambda = \infty$ but with the parameter values from Baseline and C-DN economies accordingly.

Constrained vs. Unconstrained Economies: In addition to the baseline comparison, Table 7 also contrasts the constrained and unconstrained versions of both the baseline economy (with risky intangible capital) and the C-DN economy (without risky intangible capital). The unconstrained economy is characterized by the absence of collateral constraints, allowing entrepreneurs to invest in any amount of capital regardless of their available financial assets. Therefore, comparing column one to column three, gives us a measure of the impact that borrowing constraints have on the allocations in the economy with risky intangible capital. Likewise, comparing column two to column four gives us the same measure in the economy without intangible capital.

For the baseline economy, the removal of collateral constraints leads to an increase in GDP from 0.92 to 0.97, driven by higher levels of labor (from 2.28 to 2.59) and tangible capital (from 2.29 to 2.41). This suggests that when entrepreneurs are not constrained by their financial resources, they can allocate more capital and labor to productive activities, thereby boosting overall economic output. However, the effect on intangible capital is relatively modest, with only a slight decrease from 0.57 to 0.56, indicating that the main impact of relaxing collateral constraints is on tangible rather than intangible investments. The total factor productivity (TFP) sees a small decline, indicating that while output increases, the efficiency of production may decrease slightly due to the expansion of less productive firms that can now access more capital. The wage bill to GDP ratio decreases slightly, while the profits to GDP ratio increases, reflecting a shift in income

Table 8: The Role of the Collateral Constraint: Inequality

	Baseline	C-DN Economy	Baseline Unconstrained	C-DN Economy Unconstrained
Gini Wealth	0.67	0.5	0.7	0.56
Gini Income	0.43	0.31	0.36	0.29
Gini Consumption	0.15	0.08	0.15	0.07
Consumption Ratio E/W	8.18	5.89	9.21	6.75
Hours Ratio E/W	0.15	N/A	0.11	N/A
Std Consumption Ent.	3.8	1.14	4.25	0.99
Std Consumption Workers	0.78	0.12	0.28	0.07
Std Hours Ent.	0.003	N/A	0.003	N/A
Std Hours Workers	0.08	N/A	0.04	N/A

Notes: C-DN Economy denotes Cagetti, De-Nardi economy i.e. economy without risky intangible capital following [Cagetti and De Nardi \(2006\)](#) setup. Unconstrained economies are counterfactuals with no collateral constraints for the entrepreneurs, i.e. $\lambda = \infty$ but with the parameter values from Baseline and C-DN economies accordingly.

towards capital owners when constraints are lifted. Interest income as a share of GDP remains relatively stable.

In the C-DN economy, the impact of removing collateral constraints is even more pronounced. GDP increases significantly from 0.71 to 0.79, with substantial increases in both labor (from 1.7 to 1.79) and tangible capital (from 1.55 to 1.88). The absence of intangible capital in this economy means that the entire increase in output is driven by greater tangible investments and labor input. Interestingly, TFP increases slightly in the unconstrained C-DN economy, indicating a more efficient allocation of resources when entrepreneurs are not financially constrained. The wage bill to GDP ratio remains stable, while the profits to GDP ratio decreases slightly, suggesting a more balanced distribution of income between labor and capital in this scenario. The interest income to GDP ratio decreases, reflecting the reduced reliance on borrowing when collateral constraints are removed.

These comparisons highlight the significant role that collateral constraints play in limiting economic potential, particularly in the allocation of tangible capital and labor. The differences between the baseline and C-DN economies also underscore the importance of intangible capital in shaping the dynamics of resource allocation and productivity in the presence of financial frictions.

Table 8 presents a comparison of wealth, income, and consumption inequality metrics across four different economic scenarios: the baseline constrained economy with risky intangible capital, the baseline unconstrained economy, the C-DN constrained economy without intangible capital, and the C-DN unconstrained economy. The table provides key inequality indicators such as the Gini coefficients for wealth, income, and consumption, the consumption ratio of entrepreneurs to workers, the standard deviation of consumption and hours for both entrepreneurs and workers, and the hours ratio of entrepreneurs to workers.

In the baseline economy, relaxing collateral constraints (moving from the constrained to the unconstrained scenario) leads to an increase in wealth inequality, with the Gini coefficient rising from 0.67 to 0.7. Similarly, income inequality increases as indicated by the Gini coefficient moving from 0.43 to 0.36. The standard deviation of entrepreneurial consumption also increases significantly, reflecting greater disparity in consumption patterns among entrepreneurs when constraints are lifted. On the other hand, the Gini coefficient for consumption remains stable, suggesting that the increase in inequality primarily affects wealth and income rather than consumption. The consumption ratio of entrepreneurs to workers increases, indicating that entrepreneurs benefit more from the removal of constraints.

In the C-DN economy, the absence of risky intangible capital results in lower levels of inequality compared to the baseline economy. When collateral constraints are relaxed in the C-DN economy, wealth inequality (Gini coefficient) rises from 0.5 to 0.56, and income inequality increases from 0.31 to 0.29. The increase in wealth and income inequality is less pronounced than in the baseline economy, reflecting the role of intangible capital in exacerbating inequality when financial frictions are present. The consumption ratio of entrepreneurs to workers also increases, but to a lesser extent compared to the baseline economy. The standard deviation of consumption for both entrepreneurs and workers decreases in the unconstrained scenario, suggesting a more equitable distribution of consumption when collateral constraints are absent. Additionally, the hours ratio and standard deviation of hours worked by entrepreneurs remain stable, indicating that labor input is less affected by the relaxation of constraints in the C-DN economy.

8 Policy Implications

Government backed loans.

TO BE COMPLETED

Fiscal multipliers

TO BE COMPLETED

9 Conclusion

This paper provides a detailed examination of how entrepreneurial income risk influences wealth accumulation and portfolio allocation decisions among entrepreneurs. By incorporating risky intangible capital into a life-cycle model of entrepreneurship, we are able to capture key empirical regularities observed in the data, particularly the substantial financial assets that entrepreneurs hold outside their businesses. Our model demonstrates that the presence of risky intangible capital not only amplifies the savings motives of entrepreneurs but also explains the observed portfolio allocation patterns, thereby providing a more accurate depiction of entrepreneurial behavior than models lacking this feature.

Moreover, our quantitative analysis reveals the significant macroeconomic implications of financial frictions in the presence of risky intangible capital. We show that while relaxing borrowing constraints has considerable effects on aggregate output and productivity in economies without intangible capital, these effects are much more muted in economies where risky intangible capital plays a central role. This finding underscores the importance of considering the interaction between financial frictions and intangible assets when designing policies aimed at supporting entrepreneurship and economic growth. Our results also suggest that policies such as the expansion of small business loans or fiscal stimulus transfers may have varying impacts depending on the nature and riskiness of the intangible capital involved, highlighting the need for tailored approaches in economic policy-making.

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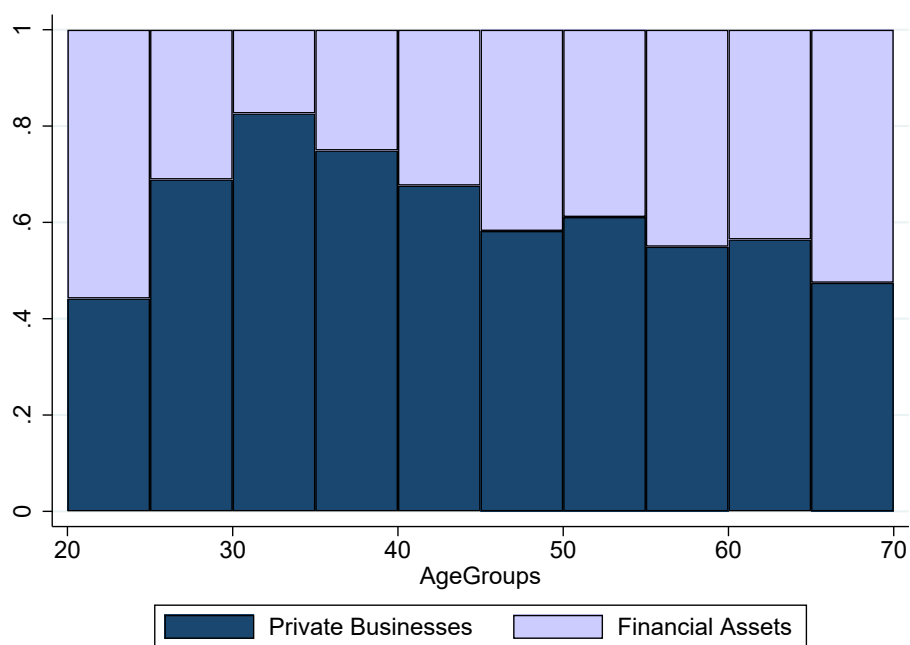
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A Additional Empirical Facts

To complement the analysis in section 3, in this section we provide additional empirical evidence from the 2019 Survey of Consumer Finances.

In figure 5, we plot the portfolio allocation of entrepreneurial wealth across 10 different 5-year age bins. We can see that financial wealth outside the business is much lower for young entrepreneurs than for older entrepreneurs.

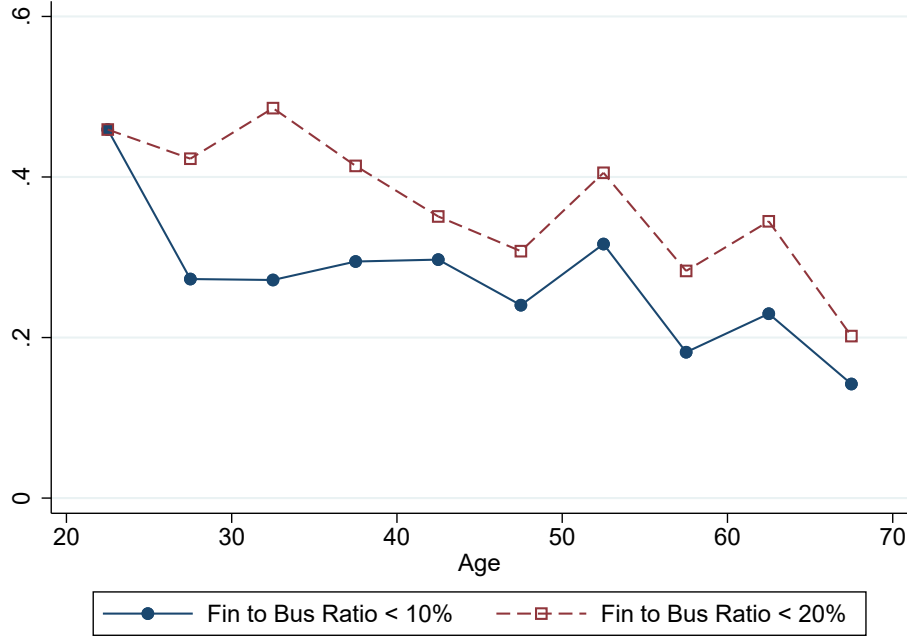
Figure 5: Portfolio Composition of Entrepreneurs over the Life-Cycle



The portfolio composition of financial assets owned by entrepreneurial households over the life-cycle. Real estate, vehicles and other tangible assets are excluded.

In figure 6, we plot the proportion of entrepreneurs who have an amount of financial assets equal to less than 10% and 20% of their business value invested outside their business. As in section 3, we interpret the financial assets held outside a business as a measure of the available resources entrepreneurs have access to. Entrepreneurs with few financial assets outside the business are most likely to be borrowing constrained.

Figure 6: Proportion of Entrepreneurs with Few Financial Assets Over the Life-cycle



The proportion of entrepreneurs with less than 10% or 20% of their business value invested in financial assets outside the business.

B Model Solution

B.1 Worker's Problem

We solve the worker's problem using the standard Endogenous Grid Method from [Carroll \(2006\)](#). We begin by re-stating the worker's backward problem, for $j = J \dots 1$

$$V_j^W(a_j, y_j) = \max_{c_j, h_j, a_{j+1}} \frac{c_j^{1-\sigma}}{1-\sigma} - \psi \frac{h_j^{1+\nu}}{1+\nu} + \beta \mathbb{E}_{y_{j+1}} [V_{j+1}^W(a_{j+1}, y_{j+1})] \quad (7)$$

subject to:

$$c_j + a_{j+1} = w y_j h_j + (1 + r^a) a_j \quad (8)$$

The problem has the following optimality conditions:

$$\frac{\partial V_j^W}{\partial c_j} = c_j^{-\sigma} - \lambda = 0 \quad (9)$$

$$\frac{\partial V_j^W}{\partial h_j} = -\psi h_j^\nu + w y_j \lambda = 0 \quad (10)$$

$$\frac{\partial V_j^W}{\partial a_{j+1}} = \beta \mathbb{E}_{y_{j+1}} \left[\frac{\partial V_{j+1}^W}{\partial a_{j+1}} \right] - \lambda = 0 \quad (11)$$

and derivative

$$\frac{\partial V_j^W}{\partial a_j} = (1 + r^a) \lambda \quad (12)$$

From this point on, we define the values of c_j , h_j and a_{j+1} that solve the worker's problem 7 as functions $c_j(a_j, y_j)$, $h_j(a_j, y_j)$ and $a_{j+1}(a_j, y_j)$. We proceed by showing how to find these.

The budget constraint 8 and conditions 9 and 10 gives us a system of non-linear equation

$$c_j = w y_j h_j + (1 + r^a) a_j - a_{j+1} \quad (13)$$

$$w y_j c^{-\sigma} = \psi h^\nu \quad (14)$$

Which, given $a_{j+1}(a_j, y_j)$ can be used to solve for $c_j(a_j, y_j)$ and $h_j(a_j, y_j)$. Finding $a_{j+1}(a_j, y_j)$ is thus done backwards in time in cases

Case 1: $j = J$: It is optimal for the worker to not save, thus

$$a_{J+1}(a_J, y_J) = 0$$

Case 2: $j = J$: The envelope theorem gives, using 12 and 9:

$$\begin{aligned}\frac{\partial V_{j+1}^W}{\partial a_{j+1}} &= \frac{\partial V_j^W}{\partial a_j} \Big|_{a_j=a_{j+1}} = (1+r^a)\lambda \Big|_{a_j=a_{j+1}} = (1+r^a)c_j(a_j, y_j) \Big|_{a_j=a_{j+1}} \\ &= (1+r^a)c_j(a_{j+1}, y_j)\end{aligned}$$

Thus, the expectation

$$\mathbb{E}_{y_{j+1}} \left[\frac{\partial V_{j+1}^W}{\partial a_{j+1}} \right] = \mathbb{E}_{y_{j+1}} \left[(1+r^a)c_{j+1}(a_{j+1}, y_{j+1}) \right]$$

which, joined with 11 and 9 yields the euler equation

$$\frac{\partial V_j^W}{\partial a_{j+1}} = \beta \mathbb{E}_{y_{j+1}} \left[\frac{\partial V_{j+1}^W}{\partial a_{j+1}} \right] = \beta \mathbb{E}_{y_{j+1}} \left[(1+r^a)c_{j+1}(a_{j+1}, y_{j+1}) \right] = \lambda = c_j^{-\sigma}$$

noting that at time j , we have access to $c_{j+1}(a_{j+1}, y_{j+1})$. We proceed by defining “future-conditioned” policies $\tilde{c}_j(a_{j+1}, y_j)$, $\tilde{h}_j(a_{j+1}, y_j)$ and our endogenous grid $\tilde{a}_j(a_{j+1}, y_j)$. From the Euler equation, we have

$$\tilde{c}_j(a_{j+1}, y_j) = \left(\beta \mathbb{E}_{y_{j+1}} \left[(1+r^a)c_{j+1}(a_{j+1}, y_{j+1}) \right] \right)^{\frac{1}{-\sigma}}$$

rearranging 14 then gives

$$\tilde{h}_j(a_{j+1}, y_j) = \left(\frac{w y_j \tilde{c}_j(a_{j+1}, y_j)^{-\sigma}}{\psi} \right)^{\nu}$$

finally, rearranging the budget constraint 8 gives our endogenous grid

$$\tilde{a}_j(a_{j+1}, y_j) = \frac{\tilde{c}_j(a_{j+1}, y_j) + a_{j+1} - w y_j \tilde{h}_j(a_{j+1}, y_j)}{(1+r^a)}$$

we then define our policy function as the first argument inverse of this endogenous grid:

$$a_j = \tilde{a}_j(a_{j+1}(a_j, y_j), y_j)$$

which is done in practice using linear interpolation.

B.2 Entrepreneur's Problem

We solve the entrepreneur's problem using a generalization of the Endogenous Grid Method for problems with higher dimensions and non-convexities from [Druehl and Jørgensen \(2017\)](#). We begin by re-stating the entrepreneur's backward problem, for $j = J \dots 1$

$$\begin{aligned} V_j^E(a_j, \kappa_j, z_j) = & \max_{c_j, h_j, q_j, a_{j+1}, \kappa_{j+1}} \frac{c_j^{1-\sigma}}{1-\sigma} - \psi \frac{h_j^{1+\nu}}{1+\nu} \\ & + \beta \left(p \mathbb{E}_{z_{j+1}} \left[V_{j+1}^E(a_{j+1}, \kappa_{j+1}, z_{j+1}) \right] + (1-p) \mathbb{E}_{z_{j+1}} \left[V_{j+1}^E(a_{j+1}, 0, z_{j+1}) \right] \right) \end{aligned} \quad (15)$$

s.t.

$$a_{j+1} + q_j + c_j = (1 + r_a) a_j + \pi(\kappa_j, z_j, a_j) \quad (\lambda_1) \quad (16)$$

$$\kappa_{j+1} = (1 - \delta_\kappa) \kappa_j + q_j^\vartheta h_j^\epsilon \quad (\lambda_2) \quad (17)$$

$$k_j \leq \lambda a_j \quad (\lambda_3)$$

$$\pi(\kappa_j, z_j, a_j) = \max_{n_j, k_j \leq \lambda a_j} z_j^{1-\gamma} \left(\kappa_j^\theta k_j^\alpha n_j^{1-\theta-\alpha} \right)^\gamma - w n_j - r_k k_j$$

$$c_j \geq 0, a_{j+1} \geq 0, \kappa_{j+1} \geq 0, q_j \geq 0, 0 \leq h_j \leq 1$$

where $r_k = r_a + \delta_k$. We define optimality conditions

$$\frac{\partial V_j^E}{\partial c_j} = c_j^{-\sigma} - \lambda_1 = 0 \quad (18)$$

$$\frac{\partial V_j^E}{\partial h_j} = -\psi h_j^\nu + \lambda_2 [\epsilon q_j^\vartheta h_j^{\epsilon-1}] = 0 \quad (19)$$

$$\frac{\partial V_j^E}{\partial q_j} = -\lambda_1 + \lambda_2 [\vartheta q_j^{\vartheta-1} h_j^\epsilon] = 0 \quad (20)$$

$$\frac{\partial V_j^E}{\partial a_{j+1}} = \beta \mathbb{E}_{z_{j+1}} \left[\frac{\partial V_{j+1}^E}{\partial a_{j+1}} \right] - \lambda_1 = 0 \quad (21)$$

$$\frac{\partial V_j^E}{\partial \kappa_{j+1}} = \beta \mathbb{E}_{z_{j+1}} \left[\frac{\partial V_{j+1}^E}{\partial \kappa_{j+1}} \right] - \lambda_2 = 0 \quad (22)$$

and derivatives

$$\frac{\partial V_j^E}{\partial a_j} = \lambda_1 \left((1+r) + \frac{\partial \pi}{\partial a_j} \right) \quad (23)$$

$$\frac{\partial V_j^E}{\partial \kappa_j} = c_j^{-\sigma} \left(\frac{\partial \pi}{\partial \kappa_j} \right) + \lambda_2 (1 - \delta_\kappa) \quad (24)$$

As before we define the values of c_j , h_j , q_j , a_{j+1} and κ_{j+1} that solve the entrepreneur's problem 15 as functions $c_j(a_j, \kappa_j, y_j)$, $h_j(a_j, \kappa_j, y_j)$, $q_j(a_j, \kappa_j, y_j)$, $a_{j+1}(a_j, \kappa_j, y_j)$ and $\kappa_{j+1}(a_j, \kappa_j, y_j)$ and proceed by showing how to find these.

Joining 23 and 18 into one equation, and 24 and 19 into another we get

$$\frac{\partial V_j^E}{\partial a_j} = c_j^{-\sigma} \left((1+r) + \frac{\partial \pi}{\partial a_j} \right) \quad (25)$$

$$\frac{\partial V_j^E}{\partial \kappa_j} = c_j^{-\sigma} \left(\frac{\partial \pi}{\partial \kappa_j} \right) + \frac{\psi h_j^{1+\nu-\epsilon}}{\epsilon q_j^\vartheta} (1 - \delta_\kappa) \quad (26)$$

As in the worker's problem, we now define our “future-conditioned” policies $\tilde{c}_j(a_{j+1}, \kappa_{j+1}, z_j)$, $\tilde{h}_j(a_{j+1}, \kappa_{j+1}, z_j)$ and $\tilde{q}_j(a_{j+1}, \kappa_{j+1}, z_j)$ as well as our endogenous grids $\tilde{a}_j(a_{j+1}, \kappa_{j+1}, z_j)$ and $\tilde{\kappa}_j(a_{j+1}, \kappa_{j+1}, z_j)$.

Joining 18 and 21 gives us

$$\tilde{c}_j(a_{j+1}, \kappa_{j+1}, z_j) = \left(\beta \mathbb{E}_{z_{j+1}} \left[\frac{\partial V_{j+1}^E}{\partial a_{j+1}} \right] \right)^{-\frac{1}{\sigma}} \quad (27)$$

Joining 19, 20, 21 and 22 gives us the system of nonlinear equations:

$$\begin{aligned} \mathbb{E} \left[\frac{\partial dV_{j+1}^E}{\partial \kappa_{j+1}} \right] \epsilon q_j^\vartheta h_j^{\epsilon-1} &= \psi h_j^\nu \\ \mathbb{E} \left[\frac{\partial dV_{j+1}^E}{\partial \kappa_{j+1}} \right] \vartheta q_j^{\vartheta-1} h_j^\epsilon &= \mathbb{E} \left[\frac{\partial dV_{j+1}^E}{\partial a_{j+1}} \right] \end{aligned}$$

which admits the following closed-form solution:

$$\begin{aligned} \tilde{q}_j(a_{j+1}, \kappa_{j+1}, z_j) &= \left(\frac{\epsilon}{\psi} \mathbb{E} \left[\frac{\partial dV_{j+1}^E}{\partial \kappa_{j+1}} \right] \left(\frac{\mathbb{E} \left[\frac{\partial dV_{j+1}^E}{\partial a_{j+1}} \right]}{\mathbb{E} \left[\frac{\partial dV_{j+1}^E}{\partial \kappa_{j+1}} \right]} \right)^{-\frac{1+\nu-\epsilon}{\epsilon}} \right)^{\frac{1}{(1-\vartheta)\frac{1+\nu-\epsilon}{\epsilon}-\vartheta}} \\ \tilde{h}_j(a_{j+1}, \kappa_{j+1}, z_j) &= \left(\frac{\mathbb{E} \left[\frac{\partial dV_{j+1}^E}{\partial a_{j+1}} \right]}{\mathbb{E} \left[\frac{\partial dV_{j+1}^E}{\partial \kappa_{j+1}} \right]} \frac{1}{\vartheta} \tilde{q}_j(a_{j+1}, \kappa_{j+1}, z_j)^{1-\vartheta} \right)^{\frac{1}{\epsilon}} \end{aligned}$$

Given these, our endogenous grid is defined by the budget constraints:

$$\tilde{a}_j(a_{j+1}, \kappa_{j+1}, z_j) = \frac{a_{j+1} + \tilde{q}_j(a_{j+1}, \kappa_{j+1}, z_j) + \tilde{c}_j(a_{j+1}, \kappa_{j+1}, z_j) - \pi(\kappa_j, z_j, \tilde{a}_j(a_{j+1}, \kappa_{j+1}, z_j))}{(1 + r_a)} \quad (28)$$

$$\tilde{\kappa}_j(a_{j+1}, \kappa_{j+1}, z_j) = \frac{\kappa_{j+1} - \tilde{q}_j(a_{j+1}, \kappa_{j+1}, z_j)^\vartheta \tilde{h}_j(a_{j+1}, \kappa_{j+1}, z_j)^\epsilon}{1 - \delta_\kappa} \quad (29)$$

where we note that 28 requires a nonlinear solver to get $\tilde{a}_j(a_{j+1}, \kappa_{j+1}, z_j)$. Finding these is done backwards in time, where, in the last period we have policy functions defined by:

- $a_{J+1}(\cdot) = 0$
- $q_J(\cdot) = h_J(\cdot) = 0 \Rightarrow \kappa_{J+1}(\cdot) = (1 - \delta_\kappa)\kappa_J$
- $c_J(\cdot) = (1 + r_a)a_J + \pi(\kappa_J, z, a)$

For $j < J$, we back-out policy functions $c_j(a_j, \kappa_j, y_j)$, $h_j(a_j, \kappa_j, y_j)$ and $q_j(a_j, \kappa_j, y_j)$ by the implicit inverses of the endogenous state space

$$\begin{aligned} c_j(a_j, \kappa_j, z_j) &= \tilde{c}_j(\hat{a}_{j+1}(a_j, \kappa_j, z_j), \hat{\kappa}_{j+1}(a_j, \kappa_j, z_j), z_j) \\ h_j(a_j, \kappa_j, z_j) &= \tilde{h}_j(\hat{a}_{j+1}(a_j, \kappa_j, z_j), \hat{\kappa}_{j+1}(a_j, \kappa_j, z_j), z_j) \\ q_j(a_j, \kappa_j, z_j) &= \tilde{q}_j(\hat{a}_{j+1}(a_j, \kappa_j, z_j), \hat{\kappa}_{j+1}(a_j, \kappa_j, z_j), z_j) \end{aligned}$$

where

$$\begin{aligned} a_j &= \tilde{a}_j(\hat{a}_{j+1}(a_j, \kappa_j, z_j), \hat{\kappa}_{j+1}(a_j, \kappa_j, z_j), z_j) \\ \kappa_j &= \tilde{\kappa}_j(\hat{a}_{j+1}(a_j, \kappa_j, z_j), \hat{\kappa}_{j+1}(a_j, \kappa_j, z_j), z_j) \end{aligned}$$

Then, for better accuracy, the policy functions $a_{j+1}(a_j, \kappa_j, y_j)$, $\kappa_{j+1}(a_j, \kappa_j, y_j)$ are derived from 16 and 17

$$\begin{aligned} a_{j+1}(a_j, \kappa_j, z_j) &= (1 + r_a)a_j + \pi(\kappa_j, z_j, a_j) - q_j(a_j, \kappa_j, z_j) - c_j(a_j, \kappa_j, z_j) \\ \kappa_{j+1}(a_j, \kappa_j, z_j) &= (1 - \delta_\kappa)\kappa_j + q_j^\vartheta(a_j, \kappa_j, z_j)h_j^\epsilon(a_j, \kappa_j, z_j) \end{aligned}$$

We describe how this is implemented numerically in Appendix B.3.

B.3 Numerical Algorithm

As per [Drue Dahl and Jørgensen \(2017\)](#), we must consider the implicit segments created by the various constraints above. Let s denote the policy at the s^{th} segment. In our problem, given that

- $c > 0$
- $a_{j+1} > 0; j < J$
- $q = 0 \Rightarrow h = 0$
- $\kappa_{j+1} = 0$ solely due to the capital shock.

we define the following three segments with pre-specified segment-specific future-dependent policy functions, which are indexed with s

- Segment 1: $q > 0, 0 < h < 1$
- Segment 2: $q = h = 0$
 - $\tilde{q}_{j,2}(\cdot) = 0$
 - $\tilde{h}_{j,2}(\cdot) = 0$
- Segment 3: $q > 0, h = 1$
 - $\tilde{h}_{j,3}(\cdot) = 1$

Given these, we get segment-specific policy functions using multi-dimensional barycentric interpolation over the endogenous a_j and κ_j grids. This routine is described in Appendix B.4. Concisely, the segment-specific policy functions $c_{j,s}(a_j, \kappa_j, y_j)$, $h_{j,s}(a_j, \kappa_j, y_j)$ and $q_{j,s}(a_j, \kappa_j, y_j)$ are defined by

$$\begin{aligned} c_{j,s}(a_j, \kappa_j, z_j) &= \tilde{c}_{j,s}(\hat{a}_{j+1,s}(a_j, \kappa_j, z_j), \hat{\kappa}_{j+1,s}(a_j, \kappa_j, z_j), z_j) \\ h_{j,s}(a_j, \kappa_j, z_j) &= \tilde{h}_{j,s}(\hat{a}_{j+1,s}(a_j, \kappa_j, z_j), \hat{\kappa}_{j+1,s}(a_j, \kappa_j, z_j), z_j) \\ q_{j,s}(a_j, \kappa_j, z_j) &= \tilde{q}_{j,s}(\hat{a}_{j+1,s}(a_j, \kappa_j, z_j), \hat{\kappa}_{j+1,s}(a_j, \kappa_j, z_j), z_j) \end{aligned}$$

where

$$\begin{aligned} a_j &= \tilde{a}_{j,s}(\hat{a}_{j+1,s}(a_j, \kappa_j, z_j), \hat{\kappa}_{j+1,s}(a_j, \kappa_j, z_j), z_j) \\ \kappa_j &= \tilde{\kappa}_{j,s}(\hat{a}_{j+1,s}(a_j, \kappa_j, z_j), \hat{\kappa}_{j+1,s}(a_j, \kappa_j, z_j), z_j) \end{aligned}$$

Then, for better accuracy, the segment-specific policy functions $a_{j+1,s}(a_j, \kappa_j, y_j)$, $\kappa_{j+1,s}(a_j, \kappa_j, y_j)$ are derived from [16](#) and [17](#)

$$\begin{aligned} a_{j+1,s}(a_j, \kappa_j, z_j) &= (1 + r_a)a_j + \pi(\kappa_j, z_j, a_j) - q_{j,s}(a_j, \kappa_j, z_j) - c_{j,s}(a_j, \kappa_j, z_j) \\ \kappa_{j+1,s}(a_j, \kappa_j, z_j) &= (1 - \delta_\kappa)\kappa_j + q_{j,s}^\vartheta(a_j, \kappa_j, z_j)h_{j,s}^\epsilon(a_j, \kappa_j, z_j) \end{aligned}$$

Finally, the actual policy functions, and by extent value function derivatives, are defined by the segment that maximizes value at any given coordinate. That is:

$$x(a_j, \kappa_j, z_j) = x_{s^*(a_j, \kappa_j, z_j)}(a_j, \kappa_j, z_j); \quad x \in (c_j, h_j, q_j, a_{j+1}, \kappa_{j+1})$$

where

$$\begin{aligned} s^*(a_j, \kappa_j, z_j) &= \arg \max_s \frac{c_{j,s}(a_j, \kappa_j, z_j)^{1-\sigma}}{1-\sigma} - \psi \frac{h_{j,s}(a_j, \kappa_j, z_j)^{1+\nu}}{1+\nu} \\ &\quad + \beta \left(p \mathbb{E}_{z_{j+1}} \left[V_{j+1}^E(a_{j+1,s}(a_j, \kappa_j, z_j), \kappa_{j+1,s}(a_j, \kappa_j, z_j), z_{j+1}) \right] \right. \\ &\quad \left. + (1-p) \mathbb{E}_{z_{j+1}} \left[V_{j+1}^E(a_{j+1,s}(a_j, \kappa_j, z_j), 0, z_{j+1}) \right] \right) \end{aligned}$$

B.4 Barycentric Interpolation Routine

In order to invert segment policies, we define two grids

- Exogenous $\mathcal{G}_{Exo} \equiv (a, \kappa)$: Fine
- Common $\mathcal{G}_{Com} \equiv (a, \kappa)$: Coarse

And for each segment, populate exogenous grid with "future-dependent" policy functions. Our objective is to interpolate from exogenous grid ($\mathcal{G}_{Exo}$) to common grid space ($\mathcal{G}_{Com}$) through the following steps:

- Divide the exogenous grid ($\mathcal{G}_{Exo}$) into triangles
 - Define each set of three adjacent points on the exogenous grid

$$\{(a'_A, \kappa'_A), (a'_B, \kappa'_B), (a'_C, \kappa'_C)\}$$

- And their implied triangles on the endogenous grid

$$\{(a_A, \kappa_A), (a_B, \kappa_B), (a_C, \kappa_C)\}$$

- For each segment ($s \in S$) and each triangle $t = \{A, B, C\} \in T_s$ within a segment
 - Where S is the number of segments
 - Where T_s is the number of triangles in segment s
- Calculate the bounding box for each triangle ($a_1^{BB}, a_2^{BB}, \kappa_1^{BB}, \kappa_2^{BB}$)

$$a_1^{BB} = \min(a_A, a_B, a_C) \times 0.90$$

$$a_2^{BB} = \max(a_A, a_B, a_C) \times 1.10$$

$$\kappa_1^{BB} = \min(\kappa_A, \kappa_B, \kappa_C) \times 0.90$$

$$\kappa_2^{BB} = \max(\kappa_A, \kappa_B, \kappa_C) \times 1.10$$

- This gives us four points that define the bounding box

$$\{(a_1^{BB}, \kappa_1^{BB}), (a_1^{BB}, \kappa_2^{BB}), (a_2^{BB}, \kappa_1^{BB}), (a_2^{BB}, \kappa_2^{BB})\}$$

- For each point on the common grid $(a_i, \kappa_i) \in \mathcal{G}_{Com}$ within the bounding box, evaluate the

three barycentric weights:

$$\begin{aligned}\omega_{i,A} &= \frac{(\kappa_B - \kappa_C)(a_i - a_C) + (a_C - a_B)(\kappa_i - \kappa_C)}{(\kappa_B - \kappa_C)(a_A - a_C) + (a_C - a_B)(\kappa_A - \kappa_C)} \\ \omega_{i,B} &= \frac{(\kappa_C - \kappa_A)(a_i - a_C) + (a_A - a_C)(\kappa_i - \kappa_C)}{(\kappa_B - \kappa_C)(a_A - a_C) + (a_C - a_B)(\kappa_A - \kappa_C)} \\ \omega_{i,C} &= 1 - \omega_{i,A} - \omega_{i,B}\end{aligned}$$

- If $\min(\omega_{i,A}, \omega_{i,B}, \omega_{i,C}) > -0.25$, use barycentric interpolate to interpolate the policy function x from the endogenous grid to the point on the common grid

$$x(a_i, \kappa_i) = \omega_{i,A} \tilde{x}(a'_A, \kappa'_A) + \omega_{i,B} \tilde{x}(a'_B, \kappa'_B) + \omega_{i,C} \tilde{x}(a'_C, \kappa'_C)$$

C Extra Tables

C.1 Exogenous Parameters

Table 9: Externally Chosen Parameters (CDN Economy)

Parameter	Value	Source
β Discount factor	0.95	Standard
σ Coefficient of relative risk-aversion	2.0	Standard
α Capital share	0.25	Standard
δ_k Depreciation of tangible capital	0.06	Standard
μ Proportion of entrepreneurs	0.11	SCF (2019)
ϑ Monetary investment to intangible capital share	0.4	Bhandari and McGrattan (2021)
ϵ Time investment to intangible capital share	0.4	Bhandari and McGrattan (2021)
θ Intangible capital to production share	0.25	Bhandari and McGrattan (2021)
δ_κ Depreciation of intangible capital	0.0	Bhandari and McGrattan (2021)

Table 10: Externally Chosen Parameters (Baseline Economy, Unconstrained)

Parameter		Value	Source
β	Discount factor	0.95	Standard
σ	Coefficient of relative risk-aversion	2.0	Standard
α	Capital share	0.25	Standard
δ_k	Depreciation of tangible capital	0.06	Standard
μ	Proportion of entrepreneurs	0.11	SCF (2019)
ϑ	Monetary investment to intangible capital share	0.4	Bhandari and McGrattan (2021)
ϵ	Time investment to intangible capital share	0.4	Bhandari and McGrattan (2021)
θ	Intangible capital to production share	0.25	Bhandari and McGrattan (2021)
δ_κ	Depreciation of intangible capital	0.0	Bhandari and McGrattan (2021)

Table 11: Externally Chosen Parameters (CDN Economy, Unconstrained)

Parameter		Value	Source
β	Discount factor	0.95	Standard
σ	Coefficient of relative risk-aversion	2.0	Standard
α	Capital share	0.25	Standard
δ_k	Depreciation of tangible capital	0.06	Standard
μ	Proportion of entrepreneurs	0.11	SCF (2019)
ϑ	Monetary investment to intangible capital share	0.4	Bhandari and McGrattan (2021)
ϵ	Time investment to intangible capital share	0.4	Bhandari and McGrattan (2021)
θ	Intangible capital to production share	0.25	Bhandari and McGrattan (2021)
δ_κ	Depreciation of intangible capital	0.0	Bhandari and McGrattan (2021)

C.2 Endogenous Parameters

Table 12: Internally Chosen Parameters (CDN Economy)

Parameter	Value
γ Returns to scale	1.0
μ_z Average entrepreneurial productivity	0.38
ψ Weight on effort disutility	1.3
ν Curvature on effort disutility	0.9
p Probability of no capital destruction shock	0.72
λ Tightness of collateral constraint	18.66
σ_w Standard deviation worker productivity shock	0.21
ρ_w Autocorrelation of worker productivity shock	1.0
σ_z Standard deviation entrepreneurial productivity shock	1.48
ρ_z Autocorrelation of entrepreneurial productivity shock	0.66
w_0^E Initial Entrepreneurial Wealth	33.96
w_0^W Initial Worker Wealth	86.89

Table 13: Internally Chosen Parameters (Baseline Economy, Unconstrained)

Parameter	Value
γ Returns to scale	1.0
μ_z Average entrepreneurial productivity	0.38
ψ Weight on effort disutility	1.3
ν Curvature on effort disutility	0.9
p Probability of no capital destruction shock	0.72
λ Tightness of collateral constraint	18.66
σ_w Standard deviation worker productivity shock	0.21
ρ_w Autocorrelation of worker productivity shock	1.0
σ_z Standard deviation entrepreneurial productivity shock	1.48
ρ_z Autocorrelation of entrepreneurial productivity shock	0.66
w_0^E Initial Entrepreneurial Wealth	33.96
w_0^W Initial Worker Wealth	86.89

Table 14: Internally Chosen Parameters (CDN Economy, Unconstrained)

Parameter	Value
γ Returns to scale	1.0
μ_z Average entrepreneurial productivity	0.38
ψ Weight on effort disutility	1.3
ν Curvature on effort disutility	0.9
p Probability of no capital destruction shock	0.72
λ Tightness of collateral constraint	18.66
σ_w Standard deviation worker productivity shock	0.21
ρ_w Autocorrelation of worker productivity shock	1.0
σ_z Standard deviation entrepreneurial productivity shock	1.48
ρ_z Autocorrelation of entrepreneurial productivity shock	0.66
w_0^E Initial Entrepreneurial Wealth	33.96
w_0^W Initial Worker Wealth	86.89

C.3 Moments

Table 15: Moments (CDN Economy)

Moments	Data	Model	Source
Worker Income Changes < 50%	0.04	0.03	DeBacker, Panousi, Ramnath (2023)
Worker Income Changes $\in [-10, 10]\%$	0.52	0.27	DeBacker, Panousi, Ramnath (2023)
Worker Income Changes > 50%	0.08	0.08	DeBacker, Panousi, Ramnath (2023)
Entrepreneur Income Changes < 50%	0.27	0.19	DeBacker, Panousi, Ramnath (2023)
Entrepreneur Income Changes $\in [-10, 10]\%$	0.14	0.12	DeBacker, Panousi, Ramnath (2023)
Entrepreneur Income Changes > 50%	0.3	0.29	DeBacker, Panousi, Ramnath (2023)
Ratio of Entrepreneur to Worker Income	3.0	5.09	SCF (2019)
Wealth to Income of Entrepreneurs	12.1	7.13	SCF (2019)
Wealth to Income of Workers	4.8	3.03	SCF (2019)
Top 1% Wealth Share	0.37	0.1	SCF (2019)
Top 10% Wealth Share	0.76	0.58	SCF (2019)
Leverage	0.46	0.47	Dinerlersoz, Kalemli-Ozcan, et al (2019)
Capital-Output Ratio	2.57	2.49	FRED (2009-2019)

Table 16: Moments (Baseline Economy, Unconstrained)

Moments	Data	Model	Source
Worker Income Changes < 50%	0.04	0.03	DeBacker, Panousi, Ramnath (2023)
Worker Income Changes $\in [-10, 10]\%$	0.52	0.27	DeBacker, Panousi, Ramnath (2023)
Worker Income Changes > 50%	0.08	0.08	DeBacker, Panousi, Ramnath (2023)
Entrepreneur Income Changes < 50%	0.27	0.19	DeBacker, Panousi, Ramnath (2023)
Entrepreneur Income Changes $\in [-10, 10]\%$	0.14	0.12	DeBacker, Panousi, Ramnath (2023)
Entrepreneur Income Changes > 50%	0.3	0.29	DeBacker, Panousi, Ramnath (2023)
Ratio of Entrepreneur to Worker Income	3.0	5.09	SCF (2019)
Wealth to Income of Entrepreneurs	12.1	7.13	SCF (2019)
Wealth to Income of Workers	4.8	3.03	SCF (2019)
Top 1% Wealth Share	0.37	0.1	SCF (2019)
Top 10% Wealth Share	0.76	0.58	SCF (2019)
Leverage	0.46	0.47	Dinerlersoz, Kalemli-Ozcan, et al (2019)
Capital-Output Ratio	2.57	2.49	FRED (2009-2019)

Table 17: Moments (CDN Economy, Unconstrained)

Moments	Data	Model	Source
Worker Income Changes < 50%	0.04	0.03	DeBacker, Panousi, Ramnath (2023)
Worker Income Changes $\in [-10, 10]\%$	0.52	0.27	DeBacker, Panousi, Ramnath (2023)
Worker Income Changes > 50%	0.08	0.08	DeBacker, Panousi, Ramnath (2023)
Entrepreneur Income Changes < 50%	0.27	0.19	DeBacker, Panousi, Ramnath (2023)
Entrepreneur Income Changes $\in [-10, 10]\%$	0.14	0.12	DeBacker, Panousi, Ramnath (2023)
Entrepreneur Income Changes > 50%	0.3	0.29	DeBacker, Panousi, Ramnath (2023)
Ratio of Entrepreneur to Worker Income	3.0	5.09	SCF (2019)
Wealth to Income of Entrepreneurs	12.1	7.13	SCF (2019)
Wealth to Income of Workers	4.8	3.03	SCF (2019)
Top 1% Wealth Share	0.37	0.1	SCF (2019)
Top 10% Wealth Share	0.76	0.58	SCF (2019)
Leverage	0.46	0.47	Dinerlersoz, Kalemli-Ozcan, et al (2019)
Capital-Output Ratio	2.57	2.49	FRED (2009-2019)